SECTION A — Very Short Answer Questions (1 Mark)

1. What is Artificial Intelligence?
2. What is Machine Learning?
3. What is Deep Learning?
4. Expand ANN.
5. Expand CNN.
6. What is data?
7. What is a feature?
8. What is a label?
9. What is labelled data?
10. What is unlabelled data?
11. What is a training dataset?
12. What is a testing dataset?
13. What is modelling?
14. What is classification?
15. What is regression?
16. What is clustering?
17. What is association rule learning?
18. What is Problem Scoping?
19. What is Data Acquisition?
20. What is Data Exploration?
21. What is Evaluation?
22. Name the 5 stages of the AI Project Cycle.
23. What is NLP?
24. What is Computer Vision?
25. Name the three domains of AI.
26. Which AI domain is used in chatbots?
27. Which AI domain is used in face recognition?
28. Give one example of supervised learning.
29. Give one example of unsupervised learning.
30. Give one application to CNN.

SECTION B — Short Answer Questions (2 Marks)

31. Differentiate between AI and ML.
32. Differentiate between ML and DL.
33. Explain features and labels with example.
34. Differentiate between labelled and unlabelled data.
35. Explain training dataset and testing dataset.
36. Explain rule-based approach.
37. Explain learning-based approach.
38. Explain classification with example.
39. Explain regression with example.
40. Explain continuous and discrete data.
41. Explain clustering with example.
42. Explain association rule with example.
43. Explain Computer Vision with example.
44. Give two applications of AI.
45. Explain Problem Scoping.
46. Explain Data Acquisition.
47. Explain Data Exploration.
48. Explain Evaluation in AI Project Cycle.
49. What is the first stage of the AI Project Cycle?
50. Name any two domains of Artificial Intelligence.
51. What is Computer Vision?
52. What is Natural Language Processing (NLP)?
53. Name the two main types of data mentioned in AI.
54. What is Bioethics?
55. What is meant by Privacy Risk in AI?
56. Name the four principles of Bioethics.

SECTION C — Long Answer Questions (3 Marks)

57. Differentiate between classification and regression.
58. Differentiate between classification and clustering.
59. Differentiate between supervised and unsupervised learning.
60. Explain the stages of AI Project Cycle.
61. Explain ANN structure.
62. Explain CNN and its applications.
63. Explain the working of a neural network.
64. Explain the role of data in AI systems.

65. Explain NLP and its applications.
66. Explain Computer Vision and its applications.
67. Explain the importance of training and testing data.
68. Explain modelling in Machine Learning.
69. Explain labelled data with suitable examples.
70. Explain association rule learning in real life.
71. Explain clustering in real life.
72. Differentiate between Computer Vision and Natural Language Processing.
73. Explain the stages of the AI Project Cycle.
74. What are the major challenges related to AI?
75. Distinguish between Sector-Based and Value-Based Ethical Frameworks.
76. Explain Rights-Based Ethical Framework with an example.

SECTION D — Case Study / Competency-Based Questions

77. A school wants to predict whether students will pass or fail using attendance and marks.
- Identify features
- Identify labels
- State whether classification or regression

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78. A mobile phone unlocks using face recognition.
- Which AI domain is used?
- Which neural network is mostly used?

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79. An online shopping app recommends products based on previous purchases.
- Which AI concept is used?
- Which domain of AI is involved?

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80. Alexa understands voice commands and replies.
- Which AI domain is used?
- Explain briefly.

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81. A company groups customers according to shopping habits.
- Which type of learning is used?

- What process is this called?

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82. A supermarket finds customers buying bread also buy butter.
- Which technique is used?
- What is this process called?

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83. A hospital predicts diabetes from patient data.
- Identify features
- Identify label
- State learning type

84. A system predicts house prices.
- Classification or Regression?
- Why?

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85. Google Translate converts English into Hindi.
- Which AI domain is used?
- Why?

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86. A self-driving car identifies traffic signs from images.
- Which AI domain is mainly used?
- Which network is commonly used?

SECTION E — 5-Mark Important Questions

87. Explain AI, ML, and DL with examples.
88. Explain the complete AI Project Cycle.
89. Explain ANN and working of neural networks.
90. Explain CNN and its applications.
91. Explain supervised and unsupervised learning.
92. Explain all three domains of AI.
93. Differentiate between NLP and Computer Vision.
94. Explain classification, regression, and clustering.
95. Explain rule-based and learning-based approaches.

ASSIGNMENT : (AI 2026 SUGGESTIONS) <1>

96. Explain the importance of data in AI systems.
97. Explain the perceptron learning model with an example .
98. Compare Rights-Based, Utility-Based, and Virtue Ethics Frameworks with suitable examples.
99. Explain the four principles of Bioethics (Autonomy, Beneficence, Non-Maleficence, and Justice) with real-life examples.